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Task Description:	This Project is aimed at testing the hypotheses that iron and proton ions of varying energies (providing high-LET values) induce mutations in exposed human cells via processes occurring in cells that undergo radiation-induced mitotic catastrophe (MC) (Swanson et al. 1995; Ianzini and Mackey 1997; Ianzini and Mackey 2007) and that this induced instability correlates with increased cancer risk. MC occurs in a variety of mammalian cell lines exposed to a broad class of agents that induce a loss of regulation of cell division and it is enhanced in cells that have diminished or that lack p53 function. p53 dysfunctions are present in more than 50% of human tumors and it is well established that germline mutations and polymorphisms in the p53 gene are present in human populations and that these populations are more radiosensitive and cancer prone than the populations not carrying these traits. It is, thus, critical to acknowledge the existence of an underestimated genetically predisposed radiosensitive human population, which might potentially represent a significant fraction of exposed personnel, and to estimate the risk for these personnel of developing cancer.		

To this end we are examining a panel of functional and not functional p53 cell lines (mouse and human) exposed to iron ions and proton beams and they response to cell killing, mutation induction, DNA damage, radiation-induced MC, and their potential for long term survival post-exposure. The aim of these studies is to determine if MC occurring after high-LET radiation of human cells is an initiating mechanism that contributes to mutagenesis and if the yield of MC can be regarded as a predictor of carcinogenesis in vivo.

Rationale for HRP Directed Research:

The studies conducted in this project fell in the category of risk #28 – Carcinogenesis - of the Radiation Health Bioastronautics Roadmap Table. In particular, we are addressing the following questions:
28b How can tissue specific probabilities for increased carcinogenesis risk from space radiation be best evaluated? What molecular, genetic, epigenetic, abscopal (effect that irradiation of a tissue has on remote non-irradiated tissue), and other factors contribute to the tissue specificity of carcinogenic risk?

To answer this question we are testing the hypothesis that high-LET radiation leads to an enhanced incidence of MC (definition and description of MC is reported in the "Task Description" section). The idea is to define MC susceptibility of human cells exposed to densely ionizing radiation to be used as a "biomarker" for delayed effects of exposure to high-LET radiations

28i What are the probabilities that space radiation will produce DNA damage at specific sites, including clustered DNA damage? What is the likelihood that DNA damage will increase the risk of carcinogenesis?

Research Impact/Earth Benefits:

To answer this question we are testing the hypothesis that DNA damage is persistent in survivors of cells that have undergone radiation-induced-MC and that mutation induction is elevated in these survivors.

28j What mechanisms modulate radiation damage at the molecular level (e.g., repair, errors in repair, signal transduction, gene amplification, bystander effects, tissue microenvironment, etc.) that significantly impact the risk of cancers, modulate the expression of radiation damage and decrease the radiation risk, and how can the understanding of mechanisms be used to predict carcinogenic risks from space radiation?

To answer this question we are following the irradiated cell populations for long time post-irradiation by means of live cell imaging. Our previous observation using the Large Scale Digital Cell Analysis System (LSDCAS) – an in-house built live cell imaging system extremely versatile – has shown that a small proportion of irradiated cells undergoing MC escape death. We are following these survivors to determine their viability. If these cells form viable colonies and are persistent for long time the risk of cancer development under these conditions can be deemed increased.

During the past year we have focused our efforts on determining the rate of cell killing, mutation induction, expression of regulators of mitosis (namely cyclin B1 and its phosphorylated form), production of DNA damage and persistence of DNA damage (delayed DNA damage), and event analyses in live cell imaging by using the Large Scale Digital Cell Analysis System (LSDCAS). Moreover, taking advantage of knowledge gained through other studies we conduct to characterize the mechanism underlying cancer resistance to radiation treatment and cancer progression [that points towards the intriguing hypothesis that cancer resistance to radiation is triggered by the activation of meiotic pathways put in place by the polyploid, multinucleated MC cancer cells to avoid death by initiating a process of depolyploidization that eventually allows these cells to gain a more stable genome composition and thus to survive] we have initiated measurements of expression of meiotic genes post-iron and proton irradiation. The aim of these studies is to determine if meiosis-specific pathways are also activated in normal cells that have undergone radiation-induced MC as a consequence of exposure to densely ionizing radiation. These studies are in line with our overall goal of establishing if survivors of radiation-induced MC remain viable, for how long they remain viable, and through which mechanism/s they remain viable. Understanding the fate of irradiated cells should allow for a better characterization of the potential carcinogenic effect of radiation exposure.
Mutated p53 MEF10(1) cells present a higher fraction of G2 cells in BrdU/PI cell cycle studies than wild-type p53 MEF12(1) cells following radiation exposure. Both MEF10(1) and MEF12(1) cells (mouse embryonic fibroblasts) undergo at least one cell division attempt post-irradiation, with a significant fraction of cells completing this initial division. The MEF10(1) cells show a greater G2 peak accumulation at 8 h post-irradiation respect to the MEF12(1) cells. Accumulation of cells in G2 phase is greater, for both cell lines, when cells are irradiated with iron ions than with gamma-rays. These data signify that cells are leaving the S/G2 cell cycle block and are entering into a premature mitosis by 8 h post-radiation. This event is further demonstrated by overaccumulation of cyclin B1 protein, a positive regulator of mitosis, during this same time frame, and it is a precondition for onset of radiation-induced mitotic catastrophe (MC). Iron ions are more effective in inducing cell killing and mutation induction in MEF10(1) and MEF12(1) cells than gamma-rays. Analysis of survival fraction reveals that iron ions are more effective in inducing cell killing than gamma-rays as it is shown by the absence of the shoulder in the low dose range and the lower overall surviving fraction at any dose in the iron ion irradiated cells compared to the gamma-irradiated cells. Equally, mutation frequency at the hprt locus increases linearly with the dose after iron ion irradiation in MEF10(1) cells while it shows a curvilinear dose-response after gamma-irradiation. Preliminary calculation of the RBE in the low dose range gives a value of about 4 for iron ions.

Persistent DNA damage is measurable in irradiated MEF12(1) cells. Preliminary data on DNA damage in irradiated MEF12(1) cells, measured as presence of phosphorylated H2AX protein, show that DNA dsb are produced post-iron ion irradiation with increases from about 5- to 20-folds over the control data. This damage is persistent during the course of the experiment (last time point for this experiment is at 7 days post-irradiation) and it correlates with live cell imaging data obtained using LSDCAS (discussed below) that show that at time of highest measurable damage MEF12(1) cells irradiated with 2 Gy of iron ions show higher probability of abnormal divisions and cell fusion, while cell death eventually peaks at 10 days post-irradiation. DNA damage data analysis for the MEF10(1) cells is still in progress.

Preliminary data with MRC-5 cells (human normal lung fibroblasts) irradiated with 1 GeV iron ions and 40 MeV proton beam. Cell cycle, mutation induction and cell killing studies are underway for this cell line. Preliminary data on cyclin B1 protein level, DNA damage and live cell imaging data analysis using LSDCAS are discussed.

The protein cyclin B1 remains at levels comparable with those from control samples in irradiated MRC-5 cells. Preliminary protein data, obtained by immunoblotting, show that cyclin B1 protein levels do not change consistently

Task Progress:

respect to the control levels in MRC-5 cells exposed to 1 GeV of iron ions. These results testify for a G1 and a G2 efficient cell cycle checkpoint control, as expected to be activated in p53 functional cell lines exposed to cytotoxic agents, and are in line with results previously published by us (Ianzini et al. 2006, 2007) that show that irradiated p53 functional cell lines do not in general present overexpression of cyclin B1 protein and are delayed for long time either in G1 or G2 of the cell cycle. The fact that these cells are not cycling, in the time frame of the experiment, is reflected in the live cell imaging data obtained using LSDCAS (discussed below) where induction of quiescence by radiation is evident for these cells.

Persistent DNA damage is measurable in irradiated MRC-5 cells. Preliminary data on DNA damage in irradiated MRC-5 cells, measured as presence of phosphorylated H2AX protein, show that DNA dsb are produced post-proton irradiation with increases from about 5- to 10-folds over the control level. This damage is persistent until day 5 and drops by day 9 post-irradiation in this experiment. More measurements are on the way to determine the overall significance of these results.

Live cell imaging carried out using LSDCAS allows for quantitative analysis of cell division, cell fusion, cell death events and for the detection of the fate of any irradiated cell in a population. During the past year great effort has been devoted to optimization, event tracking and analysis code software development for LSDCAS. LSDCAS is a live cell imaging system developed entirely in-house and capable of analyzing thousands of living cells for a period of few days up to a month in a single experiment. LSDCAS is designed to allow quantitative studies of cell populations grown under conditions identical to those used in routine biochemical/molecular investigations of a variety of phenomena. The studies conducted using LSDCAS provide quantitative measurement of the probability of events such as normal or abnormal cell division, cell fusion, induction of MC, cell death, and direct measurement of cell motility in virtually any adherent cell line, as tens of thousands of individual cells can be monitored for weeks following irradiation. This capability of following the imaged cells for such a long time also allows for the detection of rare events in the population. LSDCAS imaging data have been obtained for MEF12(1) and MRC-5 cells irradiated with 1 GeV iron ions and gamma-rays. The data were expressed as probabilities of occurrence of normal division, abnormal division, cell fusion and cell death. For the MEF12(1) cells analysis of the imaging data reveals that iron ions are much more effective than gamma-rays at inducing cell death, abnormal divisions, and cell fusion. Both abnormal division and cell fusion are hallmarks of MC. Analysis of same events in the MEF10(1) cells is still underway and it is problematic because these cells are highly motile [much more than the MEF12(1)] and often go off screen. We are developing “patching” software that will allow us to link together different imaging fields so to permit to track the cells leaving the original chosen field of vision. This development is laborious and we have not completed it yet so that the MEF10(1) imaging data analysis is not available at this moment. Based upon the molecular and biochemical data available and described above, we expect that the extent of abnormal divisions and cell fusions will be greater in the irradiated MEF10(1) cells than that reported here for the MEF12(1) cells. For the MRC-5 cells analysis of the imaging data reveals that cells irradiated with 2 Gy gamma-rays grow exponentially (essentially at the same rate as control cells) for the first 50 h post-irradiation, at which point cell division plateaus, as does the controls, probably due to cell-cell contact inhibition. Cells irradiated with iron ions appear to remain quiescent for the duration of the entire experiment, as consequence no cell division (normal or abnormal) is evident in this cell population. Cell death analysis reveals that iron ions are also more effective than gamma-rays at inducing cell killing. These data are in line with the molecular and biochemical data discussed above and provide further evidence of a functioning radiation-induced cell cycle checkpoint control in this p53 wt cell line. This withstanding, it is of note, nevertheless, the fact that the data presented here regarding the MEF12(1) cells demonstrate that these cells, although with a low yield, undergo abnormal divisions and cell fusions (both hallmarks of radiation-induced MC) upon radiation exposure, despite the fact that they possess functional p53. These data, thus, suggest that some p53 wild-type cells might be sensitive to radiation-induced MC, and, since MC is in general a property of cancer cells, substantiate the effort spent in understanding the role played by ionizing radiation in the induction of events that might be linked to radiation carcinogenesis in normal cells.

The meiosis-specific SPO11 gene is expressed in irradiated MEF12(1) cells. Other studies in our laboratory are aimed at characterizing the mechanisms underlying cancer resistance to radiation treatment and cancer progression. In these studies we have found that in p53 non functional cancer cells meiosis-specific genes are expressed during depolyploidization of polyploid (large multinucleated) cancer cells formed via radiation-induced MC, and that through the action of depolyploidization a progeny of smaller mononucleated cells (indistinguishable form the control cells) is formed that has the potential for long-lasting survival. Thus, allowing the formulation of the hypothesis that cancer cells might acquire resistance to treatment, and might as well progress, by activating pseudo-meiotic pathways. Meiotic genes expressed during meiosis prophase I and involved in chromatid pairing, synaptonemal complex formation, and chromatid cohesion (such as DMC1, SPO11, SYCP1-3, STAG3 genes) are all found activated in the study mentioned above. With the idea in mind that the same mechanisms responsible for cancer progression might also be responsible of cancer development, we have initiated measurements of expression of meiotic genes post-iron and proton irradiation of normal cells. The aim of these studies is to determine if meiosis-specific pathways are also activated in normal cells that have undergone radiation-induced MC as a consequence of exposure to densely ionizing radiation. These studies are in line with our overall goal of establishing if survivors of radiation-induced MC remain viable, for how long they remain viable, and through which mechanism/s they remain viable. Preliminary protein data, obtained by immunoblotting, show that SPO11 protein levels are increased in MEF12(1) cells post-iron ion irradiation with a peak increase between 24 and 96 h post-irradiation. SPO11 protein remains elevated up until day 11 post-irradiation that marks the end of this experiment. These are very preliminary data and the significance of them remains to be addressed, as the fact that there is a basal expression of this protein even in the control samples. This latter finding mirrors our results obtained with the cancer cells (Ianzini et al. 2008). The explanation of this finding in the cancer cell case is that the meiotic genes we find expressed can all be classified as cancer/testis antigens, and these antigens have been described in untreated cancer cells. Here, the question is why there is a measurable (although small) amount of meiotic gene/s expressed in normal control cells. An answer to this question might be found in the fact that, although these cells are considered to be normal and are p53 wt, they, nevertheless, have shown to be susceptible to radiation-induced MC. Thus, the question of how much only p53 dysfunctions are necessary for MC to occur or how much p53 polymorphism might play a role in these events remain unanswered at this point. If cells that have undergone radiation-induced MC are able to survive overcoming the detrimental effect of radiation and form viable clones, the genome integrity of these cells will de facto be compromised, and might represent a focal spot for cancer to develop. Understanding the fate of irradiated cells and their potential for long-lasting survival should allow for a better characterization of the potential carcinogenic effect of radiation exposure.

Bibliography Type:	Description: (Last Updated: 02/12/2021)
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